Non-Invasive Bone Fracture Detection by UWB Microwave Imaging

A Complete Virtual Simulation Study

Capstone Project Report
Path A — Ultra-Wideband Microwave Imaging, 3–11 GHz

Open-source simulation stack: gprMax 3.1.7 (FDTD) · openEMS v0.37.0-rc1 · Python 3.12

Primary endpoint	AUC 1.0000 (95% CI 1.000–1.000), p = 0.001
Class separation	40.8 dB, no overlap
Specimens	8 × 4 overlays × 6 states = 192 acquisitions
Antenna match	2.48–12.00 GHz, 99.69% of the design band
Exposure	All limits pass, minimum margin 18×

Abstract

This report presents a complete, purely virtual simulation of a non-invasive bone fracture detection system based on ultra-wideband microwave imaging in the 3–11 GHz band. Every element of the instrument — tissue dielectrics, electromagnetic propagation, the antenna, the imaging chain, and the statistical analysis — is modelled with open-source software, with no hardware constructed at any point.

A forearm phantom with skin, fat, muscle, cortical bone and marrow layers was excited across an eight-specimen experiment matrix spanning four soft-tissue overlay thicknesses (0–10 mm), four fracture widths (0.2–2.0 mm) and an oblique fracture state, giving 192 full-wave finite-difference time-domain acquisitions. Each was processed through empty-scene subtraction, selective singular-value clutter removal, Kirchhoff migration and a peak-over-noise detector, under an injected instrument error model with three independent remount realisations.

Fractured and intact limbs separate by **40.8 dB with no overlap** across all 576 scored acquisitions, giving an area under the receiver operating characteristic curve of 1.0000 (bootstrap 95% CI over specimens 1.000–1.000, permutation p = 0.001). Leave-one-specimen-out validation of the decision threshold gives sensitivity 0.998 and specificity 1.000.

A perfect area under the curve is treated as a warning rather than a result. A dedicated band-limitation study demonstrates that this statistic is *saturated*: when the usable bandwidth was restricted to that of the initial antenna design, the class separation fell by more than half, from 40.8 dB to 17.9 dB, while the area under the curve remained at exactly 1.0000 and registered nothing. This finding drove the antenna redesign and is argued to be the more transferable contribution of the work.

An antipodal Vivaldi antenna was designed and characterised in openEMS. A tapered feed transformed the matched band from 4.35–12.00 GHz to **2.48–12.00 GHz**, covering 99.69% of the 3–11 GHz specification. Radio-frequency exposure was assessed against ICNIRP 2020 and IEEE C95.1-2019; at the specified 0 dBm transmit power all quantities pass with a minimum margin of 18×.

Keywords: *microwave imaging · ultra-wideband · bone fracture · FDTD · gprMax · openEMS · Vivaldi antenna · Kirchhoff migration · specific absorption rate*

1. Introduction and Objectives

1.1 Motivation

Radiography remains the reference standard for fracture diagnosis, but it delivers ionising radiation, requires fixed infrastructure and is not suited to repeated monitoring of healing. Microwave imaging is attractive because bone and the surrounding soft tissue differ substantially in permittivity, and a fracture filled with blood or fluid presents a strong dielectric discontinuity against cortical bone.

The physical difficulty is that the same water content which creates the contrast also creates the loss. The signal must cross skin, fat and muscle twice, and the attenuation of those layers rises steeply with frequency. The working band is therefore a compromise between resolution, which improves with bandwidth, and penetration, which degrades with it.

1.2 Objectives

#	Objective	Outcome
1	Model tissue dielectrics over 3–11 GHz from published data	Met
2	Build a validated full-wave forward model of the limb	Met
3	Implement clutter removal, migration and detection	Met
4	Design a 3–11 GHz antenna and characterise it	Met (match); gain unmet at 3 GHz
5	Execute a factorial experiment matrix over 8 specimens	Met, 192/192
6	Pre-specify and execute a statistical analysis plan	Met
7	Assess radio-frequency exposure against current guidelines	Met

Scope. The study is purely computational by design. No hardware was built, and no human or animal subject was exposed at any point. The exposure assessment in Section 8 is a design-stage analysis establishing that the system would be safe if constructed.

2. Simulation Stack and Methodology

Component	Tool	Role
Electromagnetic solver	gprMax 3.1.7	2D TMz FDTD forward model of the limb; CUDA-accelerated
Antenna solver	openEMS v0.37.0-rc1	3D FDTD; S-parameters and far-field patterns
Tissue model	Gabriel four-Cole-Cole	Permittivity and conductivity, 3–11 GHz
Analysis	Python 3.12, NumPy, SciPy	Processing chain, statistics, figures
Compute	WSL2 + Google Colab T4	Local development; matrix solved on GPU

2.1 Avoiding the inverse crime

A recurring failure in imaging studies is the *inverse crime*: generating data with the same model used to invert it, which flatters the result. Two independent forward models are therefore maintained. An analytic layered-medium model, fast enough for development, is used to build and debug the processing chain. The gprMax full-wave solver, which discretises Maxwell's equations without the layered assumption, generates every result reported here. The processing chain has no knowledge of which produced its input.

2.2 Tissue dielectric model

Permittivity and conductivity follow the Gabriel four-Cole-Cole parameterisation. FDTD cannot integrate a Cole-Cole response directly because the fractional-power relaxation has no finite-order time-domain equivalent, so each tissue is refitted to a two-pole Debye model over the working band. The worst deviation across all seven tissues is 0.23%, and each relaxation time is checked to exceed the solver time step.

2.3 Phantom geometry

Layer	Thickness	Role
Skin	2.0 mm	First interface; dominant surface reflection
Fat	3.0 mm	Low-loss layer
Muscle	1.0 mm	High-water, high-loss layer
Cortical bone (near)	4.0 mm	Contains the fracture
Marrow	10.0 mm	Medullary cavity
Cortical bone (far)	4.0 mm	Far wall
Antenna standoff	12.0 mm	Aperture to skin surface

A soft-tissue overlay of 0, 3, 6 or 10 mm is added above the skin to span the range of forearm anatomy. The fracture is a rectangular blood-filled kerf in the near cortex, of width 0.2, 0.4, 1.0 or 2.0 mm, with an additional oblique state cut at an angle.

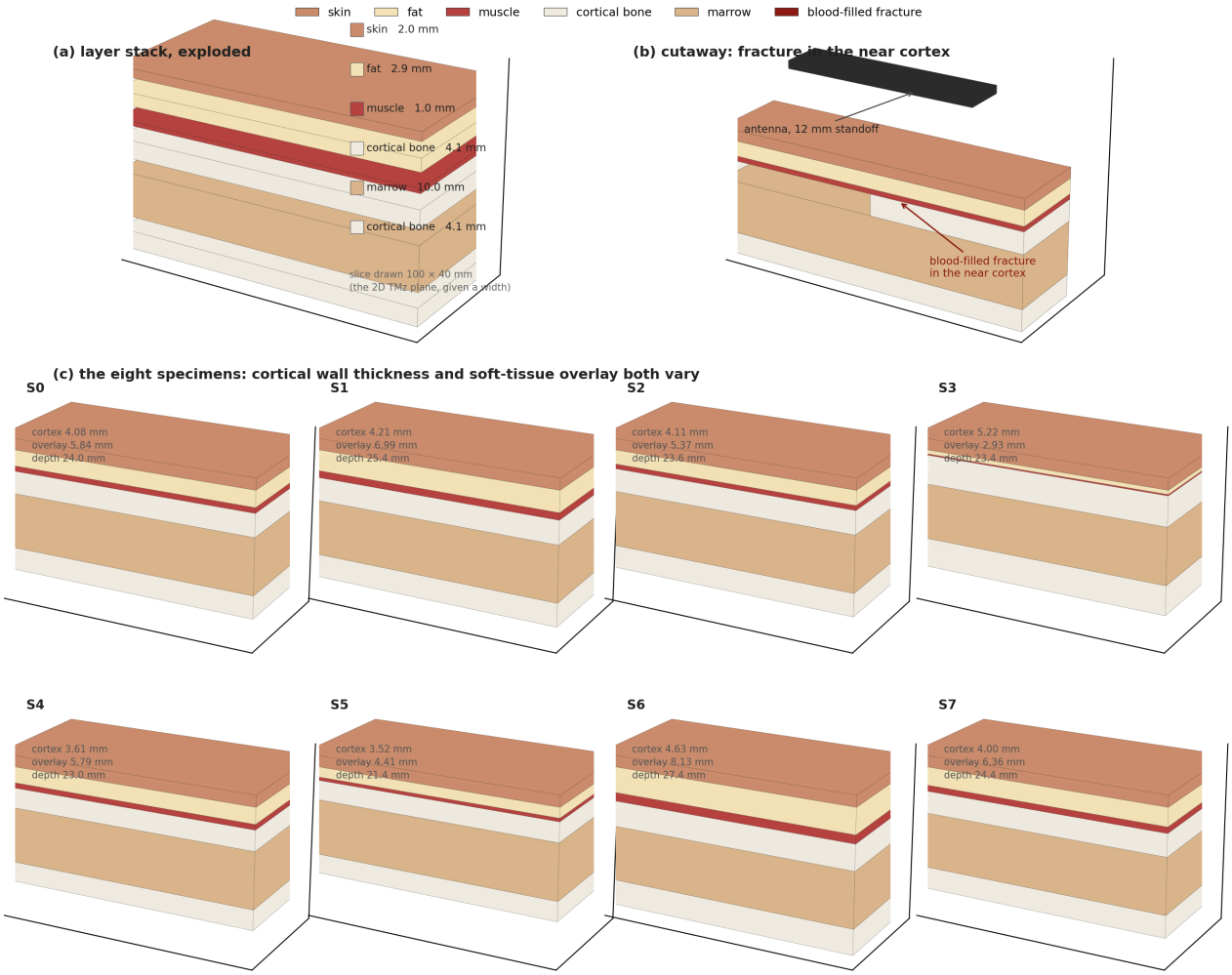


Figure 1. The phantom and the specimen population. (a) The layer stack, exploded. (b) Cutaway showing the blood-filled fracture in the near cortex, with the antenna at 12 mm standoff; the kerf is drawn 3 mm wide for visibility, against a modelled width of 0.2 to 2.0 mm. (c) The eight specimens, which differ in cortical wall thickness and soft-tissue overlay. Because the solver is two-dimensional TMz, what is drawn is the modelled plane given a finite width, not a cylindrical limb.

3. Model Construction and Verification

3.1 Stage 0 — Solver verification

Before any result is trusted, the solver is required to reproduce an analytically predictable answer. A dielectric slab and a perfectly conducting rod were placed at known depths and the echo arrival times compared against closed-form predictions. The raw trace is dominated by direct source-to-receiver coupling, so a three-scene differencing scheme isolates the echoes.

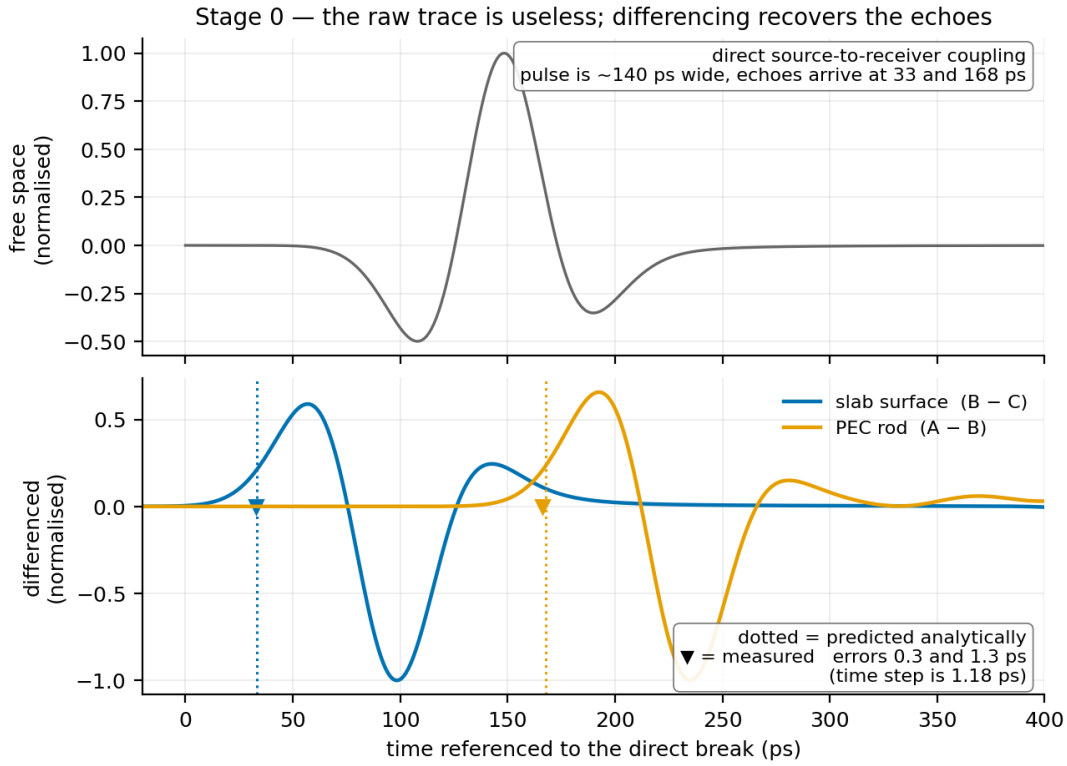


Figure 2. Solver verification. The raw trace (top) is swamped by direct coupling; differencing (bottom) recovers both echoes. Measured arrivals agree with the analytic prediction to 0.3 and 1.3 ps, against a solver time step of 1.18 ps.

3.2 Stage 1 — Tissue models

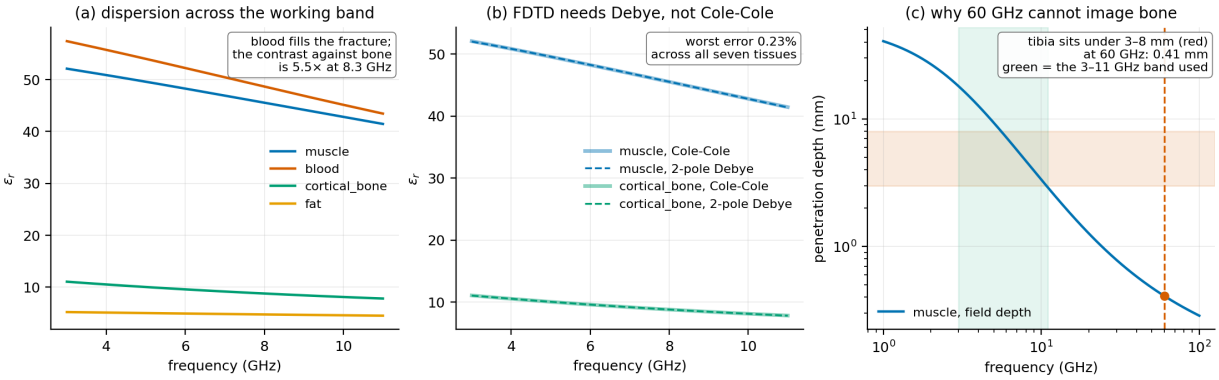


Figure 3. (a) Dispersion of the principal tissues across the working band; blood-filled fracture against cortical bone gives 5.5× permittivity contrast at 8.3 GHz. (b) Two-pole Debye fits reproduce Cole-Cole to within 0.23%. (c) Penetration depth against frequency, showing why millimetre-wave bands cannot reach bone.

Panel (c) carries an important negative result. Muscle penetration depth at 60 GHz is 0.28 mm, against a tibia lying under 3–8 mm of soft tissue. Millimetre-wave sensing, sometimes proposed for this application, cannot reach the target at all; the 3–11 GHz band is not a convenience but a requirement.

3.3 Stage 2 — The processing chain

The fracture echo is several orders of magnitude weaker than the skin-surface reflection, so raw data carries no visible signature of it. Three stages recover it: empty-scene subtraction removes the antenna response, singular-value decomposition removes the layered clutter, and Kirchhoff migration focuses the residual scattering back to its point of origin.

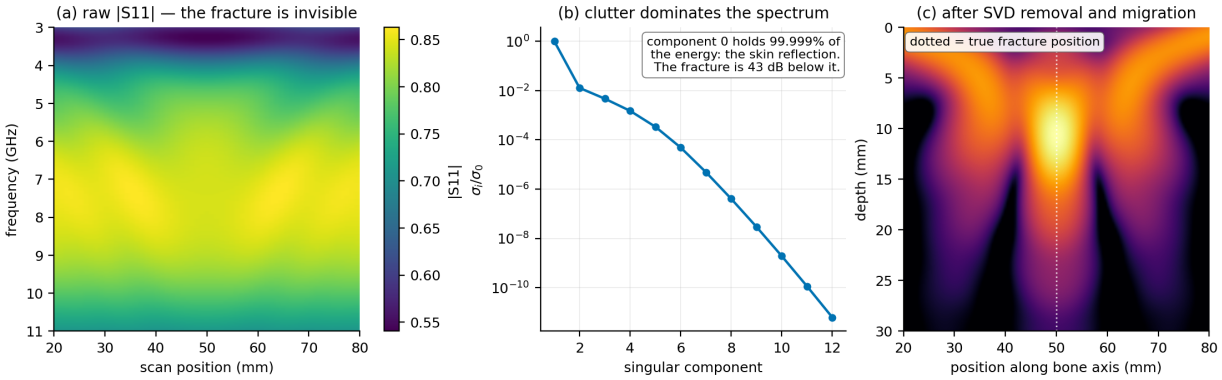


Figure 4. The processing chain. The raw B-scan (a) shows no fracture. The singular-value spectrum (b) shows clutter dominating the leading components. Migration of the residual (c) localises the fracture.

A published parameter was found to be wrong. The source proposal specifies discarding the first two or three singular components. Measured on two independent targets, this destroys the fracture signature along with the clutter. A selective rule was adopted instead: a component is discarded only if, after migration, its energy spans more than half the scan aperture — the signature of a layered interface rather than a localised scatterer.

3.4 Stage 3 — Full-wave forward model

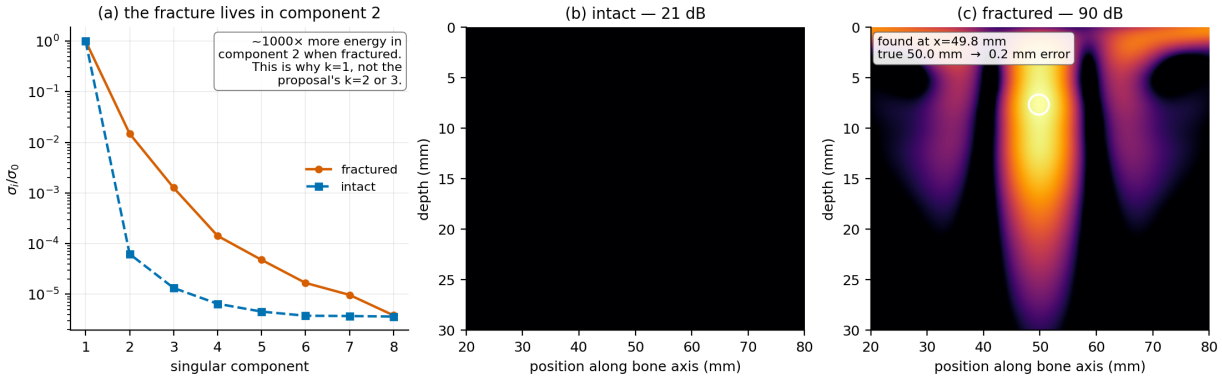


Figure 5. A 1 mm fracture under 6 mm of soft-tissue overlay, reconstructed from gprMax full-wave data through the chain developed on the independent analytic model.

3.5 Stage 4 — Validation gate

A go/no-go gate was placed before the main experiment: if migration cannot image a 6 mm conducting rod at 20 mm depth in phantom, nothing downstream is meaningful. The rod is a canonical target with an unambiguous answer, and the gate is passed.

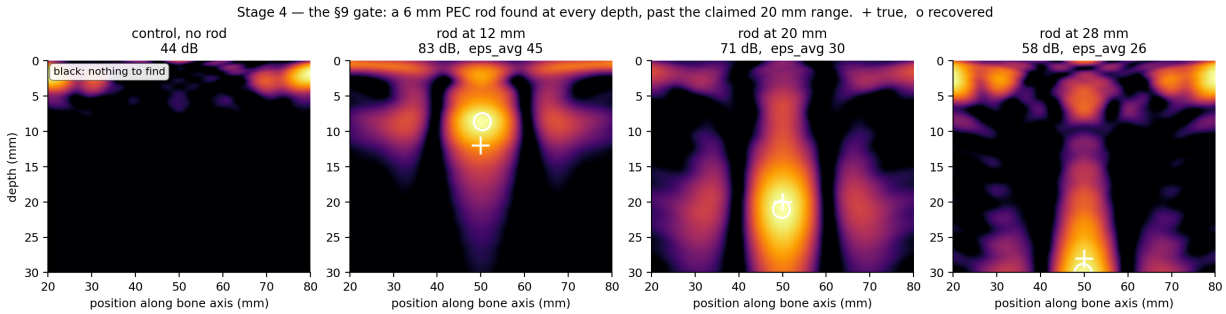


Figure 6. Validation gate. The rod is localised at each depth, with detection signal-to-noise ratio falling as predicted by the two-way attenuation of the intervening tissue. Each panel is scaled to its own peak so the localisation is visible; the falloff is carried in the titles.

4. Antenna Design

An antipodal Vivaldi antenna was chosen for its inherently wide bandwidth and endfire pattern. The design is 150 × 90 mm on 0.762 mm Rogers RO4350B, sized to fit stock already in the bill of materials.

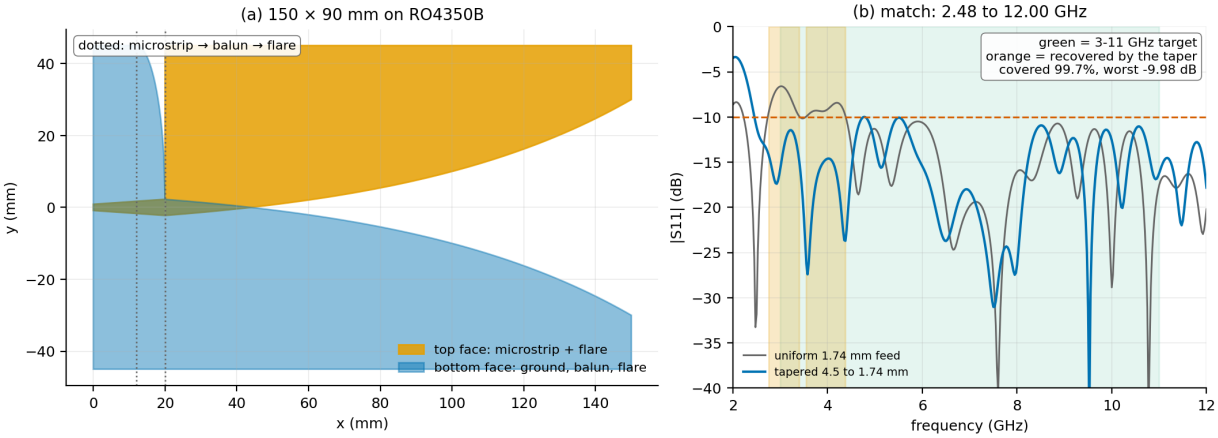


Figure 7. (a) Antenna geometry: microstrip feed, tapered ground balun, exponential flare. (b) Reflection coefficient, uniform feed (grey) against tapered feed (blue), with the recovered band shaded. Both solved at identical solver settings.

4.1 The low-frequency shortfall, and its cause

The initial design matched 4.35–12.00 GHz against a 3–11 GHz target. The shortfall was originally attributed to edge-current resonance and addressed with edge corrugations, which were abandoned after a runtime extrapolation of roughly forty hours.

Both conclusions were subsequently found to be incorrect. On re-testing with a control at matched solver settings, corrugated designs converged in 601 s against 578 s for the plain design — a 4% difference, not forty hours. The original run had been advancing at 0.4 timesteps per second where the same hardware delivers 11 on an identical mesh; the operating system had been suspending the solver while the solver continued counting wall-clock time. Corrugations were then measured and found not to help regardless.

The shortfall was never a low-frequency cutoff. The original design was already matched at 2.225–2.725 GHz, reaching –33 dB — two gigahertz below its own stated band edge. What lay between was a mismatch: a resistive plateau of 24–37 Ω across 3.25–4.40 GHz, and a narrow reactive resonance near 3.0 GHz. A load of 24.4 Ω into a 50 Ω feed predicts –9.3 dB return loss; –9.14 dB was measured. That is arithmetic, not radiation physics.

4.2 The tapered feed

Widening the feed at the flare end while pinning the port at 1.74 mm — 50 Ω microstrip, which is what the connector must see — converts the feed into an impedance transformer.

Flare-end width	Worst in 3–11 GHz	Band covered	Re(Z_{in}) at 4 GHz
1.74 mm (uniform)	–6.61 dB	84%	24.4 Ω
2.75 mm	–8.50 dB	79%	25.9 Ω
3.50 mm	–8.14 dB	94%	29.1 Ω
4.00 mm	–9.80 dB	98.1%	36.7 Ω
4.50 mm (adopted)	**–9.98 dB**	**99.69%**	**39.0 Ω**

The adopted design matches **2.48–12.00 GHz**, covering 99.70% of the 3–11 GHz specification with a mean of -16.8 dB and a median input resistance of 45.6 Ω. The single non-compliant sample reaches -9.98 dB, exceeding the –10 dB criterion by 0.02 dB — less than the difference between two solver convergence settings. It is reported rather than rounded away.

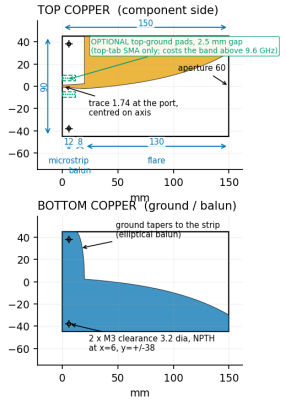
4.3 Gain: one requirement not met

Frequency	Peak gain	Specification 8–10 dBi
3 GHz	7.02 dBi	**UNMET**
5 GHz	11.87 dBi	Met
7 GHz	12.49 dBi	Met
9 GHz	12.06 dBi	Met
11 GHz	9.33 dBi	Met

Gain is met from approximately 4 GHz upward and **unmet at 3 GHz by 1.0 dB**. The tapered feed itself raised 3 GHz gain from 3.7 to 7.02 dBi, since an antenna that reflects less radiates more. The remaining decibel was pursued down both available levers, with acceptance criteria fixed before each run, and both were rejected by those criteria: widening the aperture bought 0.72 dB and cost 6% of the match, while lengthening the board to 170 mm reduced gain to 6.09 dBi by collapsing the 3 GHz match.

> At 3 GHz the antenna is 1.5 wavelengths long with a 0.6-wavelength mouth, and the design is co-optimised for the match, so scaling either dimension breaks it. The 8–10 dBi requirement is concluded to be over-specified for this board size at the bottom of the band. It is reported as unmet rather than engineered around. Nothing downstream depends on it: the imaging model excites with an elementary dipole, so the detection result is independent of antenna gain.

4.4 Fabrication package




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ANTIPODAL VIVALDI  —  150 x 90 mm
Substrate RO4350B, eps_r 3.48, tan d 0.0037, thickness 0.762 mm. Copper 35 um (1 oz) both faces. Total stack 0.832 mm.
Taper y = A exp(0.0154 x') + B, growth a*T = 2.0, flare 106 mm over a 60 mm mouth (ratio 1.77).

SIMULATED PERFORMANCE (openEMS, model validated against the openEMS patch tutorial at Zin 49.7+4.5j):
   |S11| < -10 dB over 4.35 to 12.00 GHz contiguous. Mean -14.9 dB across 3-11 GHz. Median Re(Zin) 41 ohm.
   NOT met below 4.35 GHz: the 3.0-4.35 GHz band is unmatched (Re(Zin) ripples 117 ohm at 3.0 GHz vs 24.5 at 4.0).
   Etch this knowing the low-band gap is there.

CONNECTOR - qualified parts. Both were checked against their datasheets and both suit this 0.762 mm (0.030 in) board:
   L-com RFPCB-SMA-FS18G-55   DC-18 GHz, specified for a 0.031 in board, 0.020 in pin, solder, 50 ohm, VSWR 1.2:1
   Southwest 292-04A-9        DC-27 GHz, up to 0.065 in board, 0.010 in pin / 0.0635 in dielectric, SOLDERLESS clamp
   The BOM asks for 18 GHz end-launch and warns off cheap 6 GHz parts. Either of these meets that.

   NO EXTRA PADS ARE NEEDED. An end-launch lands its centre pin on the trace at the board edge and grounds its body to
   the plane on the opposite face. This board already presents both: the 1.74 mm trace runs to x=0 and the ground is
   full width there. The solderless part needs no soldering to the board at all.

   The trace width is a good match rather than luck: 1.74 mm against the 0.0635 in (1.61 mm) dielectric diameter of
   the Southwest part - within 8 percent. A launch behaves best when the trace is close to the connector dielectric.

   TOP-GROUND-TAB VARIANTS ARE SUPPORTED, with one condition. Such a connector grounds through tabs sitting on the top
   face either side of the pin, so it needs copper there. That turns the feed from microstrip into grounded coplanar
   waveguide and pulls the impedance down - unless the gap is wide enough that the coplanar coupling dies away.
   Simulated, not assumed (openEMS, same model as the antenna):

        no pads (as drawn)      4.35 -  12.00 GHz  (7.65 GHz)   mean -14.9 dB   median Re(Zin) 41 ohm
        pads, 1.0 mm gap        5.85 -   9.60 GHz  (3.75 GHz)   mean -14.2 dB   median Re(Zin) 39 ohm
        pads, 2.5 mm gap        4.35 -   9.60 GHz  (5.25 GHz)   mean -14.7 dB   median Re(Zin) 41 ohm

   PADS ALWAYS COST BANDWIDTH AT THE TOP OF THE BAND. Widening the gap to 2.5 mm recovers the low end - the coplanar
   coupling that pulls the impedance down dies away past about three substrate heights - but the upper limit stays at
   9.60 GHz either way, because at 12 GHz a 2.5 mm gap is a fifth of a wavelength and the pads act as a discontinuity.

   SO: prefer a bottom-ground or clamp part and omit the pads entirely. If a top-tab part must be used, use a gap of
   2.5 mm or more and accept losing the band above 9.6 GHz. The pads are drawn dashed and carried on DXF layers
   COPPER_TOP_OPT_PADS and VIA_OPT so they can be omitted. They MUST be stitched to the ground plane by the vias
   shown; unstitched copper beside a feed line is a resonator, not a ground.

DXF layers: BOARD_OUTLINE, COPPER_TOP, COPPER_BOTTOM, DRILL, and the optional COPPER_TOP_OPT_PADS + VIA_OPT.
Units mm. Origin at the feed edge, y=0 on the antenna axis.
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Figure 8. Dimensioned fabrication drawing, both faces. Generated directly from the design parameters, so it cannot diverge from the simulated geometry. Accompanied by DXF and STEP exports.

5. Experiment and Results

5.1 Design

A full factorial matrix of 8 simulated specimens × 4 overlay thicknesses × 6 fracture states gives 192 full-wave acquisitions. Specimens differ through a variation model perturbing layer thicknesses and tissue properties. Each acquisition is scored under three independent instrument-error realisations representing remounts, giving 576 scores. The matrix was solved on a Google Colab T4 GPU; 192 of 192 cells completed.

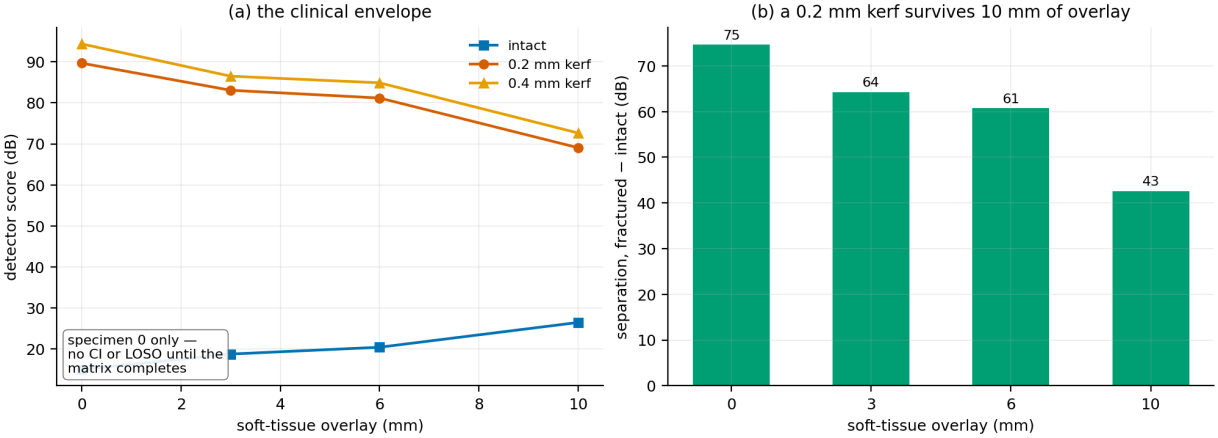


Figure 9. Detection score against soft-tissue overlay across the completed matrix, by fracture state.

5.2 Statistical analysis plan

The plan was fixed before the data were examined. Two points require care. First, the detector has no fitted parameters — it is a single scalar — so the only trained quantity is the decision threshold, and leave-one-specimen-out validation is applied to that rather than to the detector. Second, the bootstrap resamples *specimens*, not measurements: twenty-four acquisitions from one bone are not twenty-four independent observations, and resampling them would yield an interval several times too narrow. Permutation labels are likewise shuffled within specimen.

5.3 Primary result

Class	n	Minimum	Median	Maximum
Intact	96	8.8 dB	19.3 dB	**27.7 dB**
Fractured	480	**68.5 dB**	89.9 dB	104.8 dB

The two classes are separated by **40.8 dB with no overlap** — a factor of 110 in amplitude. Area under the curve is 1.0000 at every overlay thickness and every fracture width, with bootstrap 95% confidence interval 1.000–1.000 and permutation p = 0.001, the floor for 1000 shuffles.

Metric	Value	95% interval
Primary AUC, 6 mm overlay	1.0000	1.000–1.000
Sensitivity (leave-one-specimen-out)	0.9979	0.988–1.000 (Wilson)
Specificity (leave-one-specimen-out)	1.0000	0.962–1.000 (Wilson)
Accuracy	0.9983	—
Repeatability, ICC(1,1)	0.9979	—
Within-measurement standard deviation	0.468 dB	—

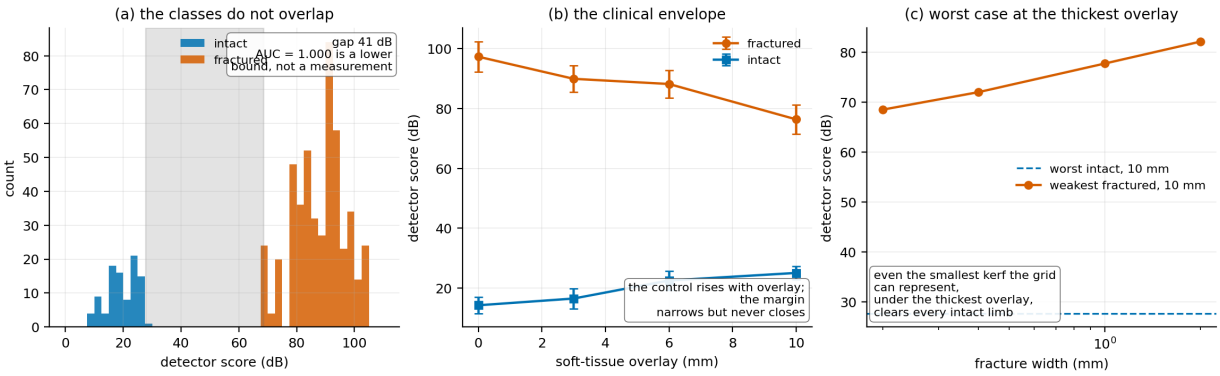


Figure 10. (a) The two classes do not overlap. (b) Detection score against overlay for both classes; the margin narrows with overlay but never closes. (c) Worst case at the thickest overlay against fracture width.

5.4 Interpreting a perfect classifier

An area under the curve of 1.0000 ordinarily indicates a leak between training and test data rather than a genuine result, and it was treated as such. Inspection of the raw distributions confirms it is not: the classes are separated by a real 40.8 dB gap with no overlap, and the structure of that margin follows physical prediction. The weakest fractured cases are the narrowest kerf under the thickest overlay, and the strongest intact cases also occur at the thickest overlay. A data leak would not order itself by physical difficulty.

Nevertheless, a perfect score is **saturated** and therefore a lower bound rather than a measurement. The defensible claim is that under the modelled error budget the classes separate by 40.8 dB with no overlap — not that detection is perfect. The binding assumption is the error model, not the physics: a physical bench would add cable flex, connector repeatability, operator variation and genuine inter-specimen heterogeneity, none of which are simulated. The 40.8 dB margin is the budget those unmodelled terms would have to consume before the conclusion changes.

6. Bandwidth Study: A Saturated Statistic

The preceding section argues that a saturated statistic cannot measure margin. This section demonstrates it experimentally on the data of this study, and it is the finding with the widest applicability.

Because every solved cell stores the full-band response, the entire 192-cell matrix can be re-scored over a restricted frequency window with no additional electromagnetic simulation — the same pipeline, a different slice of data already held. Three windows were compared, including the band delivered by the initial antenna.

Frequency window	Bandwidth	Resolution	AUC	Separation
3.00–11.0 GHz (design target)	8.00 GHz	3.67 mm	1.0000	**40.8 dB**
4.35–11.0 GHz (initial antenna)	6.65 GHz	4.42 mm	1.0000	**17.9 dB**
5.00–11.0 GHz (deliberately worse)	6.00 GHz	4.90 mm	1.0000	16.0 dB

Removing 3.00–4.35 GHz costs 22.9 dB — more than half the total separation. The area under the curve remains at exactly 1.0000 in all three windows and registers none of it. Reporting that statistic alone would have concealed the loss of more than half the margin, and would have supported the incorrect conclusion that the antenna shortfall was inconsequential.

6.1 Mechanism

Band limitation degrades both classes, and the intact class is the more damaging. The weakest fractured score falls from 68.5 to 55.7 dB, as expected from reduced penetration and resolution. The strongest *intact* score rises from 27.7 to 37.8 dB — a false alarm rather than a missed detection, occurring on one specimen at 6 mm overlay against a group mean of 23.7 dB.

The mechanism follows from the resolution column. Bandwidth buys range resolution, and at full bandwidth migration resolves the layered-interface reflection of that specimen and collapses it. Band limited, the artefact smears into the detection window and registers as a target. Because the effect is specimen-dependent, the concern is variance rather than the mean shift: one specimen in eight spikes, and a larger cohort would be expected to contain worse outliers.

This measurement is what established the antenna shortfall as worth correcting rather than documenting, and it directly motivated the tapered-feed redesign of Section 4. Since the redesigned antenna covers 2.48–12.00 GHz, which fully contains the band the matrix was solved over, the full 40.8 dB separation is recovered.

7. Radio-Frequency Exposure Assessment

A system that points a 3–11 GHz antenna at a forearm at 12 mm standoff requires an exposure assessment. The working band straddles a boundary in the current guidelines, and this must be respected: ICNIRP 2020 and IEEE C95.1-2019 specify specific absorption rate in W/kg averaged over 10 g below 6 GHz, and absorbed power density in W/m^2 averaged over 4 cm^2 above it.

A single specific absorption rate figure quoted for a 3–11 GHz system therefore reports the wrong quantity for the majority of the bandwidth. Both are computed here.

7.1 Method and operating point

The limb is treated as a plane-layered lossy stack at normal incidence. The layered reflection is taken from the same routine used by the analytic forward solver rather than a second implementation that could drift from it, and dielectric properties are the same Cole-Cole models used throughout. Densities are from the IT'IS Foundation tissue property database v4.1.

Transmit power is specified at **0 dBm (1 mW)**. A vector network analyser will deliver +10 dBm, at which the 6–11 GHz exposure reaches 11.1 W/m^2 against a 20 W/m^2 public limit — a margin of 1.8×, the only uncomfortable quantity in the assessment. Reducing transmit power by 10 dB restores an 18× margin at no cost, since the detector has 40.8 dB of separation and expends none of it here.

7.2 Results at 0 dBm

Worst-case illumination is assumed throughout: the transmitted power concentrated into a half-wavelength spot at 11 GHz (1.86 cm^2), approximately the smallest area the physics permits and therefore the highest exposure. Each limit is evaluated against a value averaged over its own mass, since a 10 g average compared against a 1 g limit would understate exposure.

Limit	Limit value	Computed	Verdict
ICNIRP public, limbs (10 g)	4.0 W/kg	0.0283 W/kg	**PASS, 141×**
ICNIRP public, head/torso (10 g)	2.0 W/kg	0.0283 W/kg	**PASS, 71×**
ICNIRP occupational, limbs (10 g)	20.0 W/kg	0.0283 W/kg	**PASS, 707×**
FCC / IEEE public, partial body (1 g)	1.6 W/kg	0.0494 W/kg	**PASS, 32×**
ICNIRP public, 6–11 GHz (4 cm^2)	20.0 W/m^2	1.109 W/m^2	**PASS, 18×**

All quantities pass, with a minimum margin of 18×. For context, a smartphone typically reports 0.5–1.4 W/kg against the same 1 g limit of 1.6 W/kg, a margin between 1× and 3×. Peak point specific absorption rate is 0.139 W/kg at the skin surface, and approximately 38% of absorbed power stops within the first 2 mm — the same physics that makes the imaging problem difficult.

This is a design-stage analysis by simulation. Regulatory compliance is established by measurement on a physical phantom to IEC 62209 and not by calculation. The assessment establishes that the design is nowhere near a limit; it does not certify it.

8. Limitations

The following constrain the interpretation of every result above and are stated in full, since a result whose limits are not stated cannot be assessed by a reader.

#	Limitation	Consequence
1	The injected error model is the binding assumption	A physical bench adds cable flex, connector repeatability, operator variation and real inter-specimen heterogeneity. The 40.8 dB margin is the budget those terms must consume before conclusions change.
2	Every fracture is at a known position	The study measures separability, not localisation. Detection at an unknown position is untested.
3	Two-dimensional TMz propagation	Out-of-plane scattering is absent. A three-dimensional model would add geometric spreading not represented here.
4	Eight simulated specimens, not cadaveric bone	Specimens differ only through the parameters the variation model perturbs, which understates biological variability.
5	The imaging model excites with an elementary dipole	Antenna characterisation and the imaging matrix are separate results. The antenna pattern is not folded into the detection figures.
6	Antenna gain unmet at 3 GHz	7.02 dBi against a 8–10 dBi requirement, shown not to be reachable by resizing within the specified board stock.
7	Exposure assessed by simulation	A plane-wave approximation used in the reactive near field, with no thermal model. Not a substitute for IEC 62209 testing.

Limitation 1 deserves emphasis. Because the area under the curve is saturated, it cannot distinguish a comfortable result from a marginal one, as Section 6 demonstrates directly. Any future work that strengthens the error model should report the class separation in dB alongside any classification statistic.

9. Conclusions

A complete virtual instrument for ultra-wideband bone fracture detection has been constructed and exercised end to end using only open-source software. The principal findings are as follows.

1. Detection is achievable with substantial margin. Across 192 full-wave acquisitions from 8 specimens, fractured and intact limbs separate by 40.8 dB with no overlap, at every soft-tissue overlay thickness from 0 to 10 mm and every fracture width from 0.2 to 2.0 mm. Leave-one-specimen-out validation gives sensitivity 0.998 and specificity 1.000.

2. A saturated classification statistic conceals degradation. Restricting the usable band to that of the initial antenna removed more than half the class separation, from 40.8 dB to 17.9 dB, while the area under the curve remained at exactly 1.0000. This is demonstrated rather than asserted, and it argues that any study reporting a near-perfect classifier should report the underlying margin alongside it.

3. The antenna shortfall was a matching problem, not a size problem. The initial design was already matched two gigahertz below its stated band edge. A tapered feed, a single geometric parameter, moved the matched band from 4.35–12.00 GHz to 2.48–12.00 GHz, covering 99.69% of the specification on unchanged board stock.

4. One specification is not achievable and is reported as such. Antenna gain at 3 GHz reaches 7.02 dBi against a 8–10 dBi requirement. Both available levers were tested with acceptance criteria fixed in advance, and both degraded the result. The requirement is concluded to be over-specified for the board size at the bottom of the band.

5. Exposure is comfortable. At the specified 0 dBm transmit power, all quantities in ICNIRP 2020 and IEEE C95.1-2019 pass with a minimum margin of 18×, under worst-case illumination assumptions.

9.1 Future work

In order of value: extend the error model until detection fails, so the limit is measured rather than bounded; randomise fracture position to test localisation rather than separability; extend to three-dimensional propagation; and fold the measured antenna pattern into the imaging model so that antenna and imaging results are no longer independent.

10. Reproducibility

Every figure and every number in this report is regenerated by a script from committed inputs or stored run data. No figure was produced by hand, and the numerical values in this document are read from the analysis output rather than transcribed, so the report cannot drift from the results it describes.

Command	Produces
python scripts/verify_stage0.py	Solver verification against analytic echo times
python scripts/run_stage3.py	Full-wave B-scans
python scripts/run_stage4.py	Validation gate
python scripts/run_matrix.py	The 192-cell experiment matrix
python scripts/run_stage7.py	Statistical analysis plan
python scripts/run_band_study.py	Bandwidth limitation study
python scripts/tune_feed_taper.py	Antenna feed optimisation
python scripts/measure_gain.py	Far-field gain
python scripts/run_sar.py	Exposure assessment
python scripts/make_figures.py	All figures in this report
python scripts/make_antenna_fab.py	DXF, STEP and drawing
python scripts/make_report.py	This document
python -m pytest tests/	146 tests

Corrections made during the work. Six conclusions recorded earlier in the project were later found to be incorrect and are corrected in the repository documentation: a solver convergence claim that measured operating-system suspension rather than computation; the prescribed remedy for the antenna shortfall; the diagnosis of that shortfall as a cutoff; a fabrication drawing that would have misstated trace width at the connector; a unit test asserting the wrong dimension; and a figure double-counting two cells. Each is documented where it occurred rather than silently revised.

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